SIMATS ENGINEERING

SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Course: Fundamentals of Data Science

Course Code: DSA0404

Slot: A

Faculty: KRISHNARAJ M

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Problem Solving Using Numpy, Pandas, Scipy, Scikit Learn Libraries in Python for Data Scientist

1. GENERATE RAW MOCK DATA (Simulating an imperfect CSV file)

Code:

File Edit Format Run Options Window Help

```
import pandas as pd
import numpy as np

# 1. GENERATE RAW MOCK DATA (Simulating an imperfect CSV file)
raw_data = {
    "Employee_Name": [" Alice ", "Bob", "Charlie", "Diana", "Evan"],
    "Department": ["HR", "Engineering", "HR", "Engineering", "Marketing"],
    "Salary": [50000, 85000, np.nan, 92000, 60000],  # Missing value
    "Join_Date": [
        "2022-01-15",
        "2021-06-20",
        "2023-03-11",
        "2020-11-01",
        "2024-02-28",
    ],
}

# Create DataFrame
df = pd.DataFrame(raw_data)

# Display Raw Data
print("--- Raw Data ---")
print(df)
```

Output:

```
--- Raw Data ---
  Employee_Name   Department   Salary   Join_Date
0         Alice           HR  50000.0  2022-01-15
1           Bob  Engineering  85000.0  2021-06-20
2       Charlie           HR      NaN  2023-03-11
3         Diana  Engineering  92000.0  2020-11-01
4          Evan    Marketing  60000.0  2024-02-28
>>>
```

2. DATA PREPROCESSING

Code:

```
File  Edit  Format  Run  Options  Window  Help
import pandas as pd
import numpy as np

# Employee Data
raw_data = {
    "Employee_Name": [" Alice ", "Bob", "Charlie", "Diana", "Evan"],
    "Department": ["HR", "Engineering", "HR", "Engineering", "Marketing"],
    "Salary": [50000, 85000, np.nan, 92000, 60000],
    "Join_Date": [
        "2022-01-15",
        "2021-06-20",
        "2023-03-11",
        "2020-11-01",
        "2024-02-28"
    ]
}

df = pd.DataFrame(raw_data)

# 2. DATA PREPROCESSING

# Remove extra spaces from Employee_Name
df["Employee_Name"] = df["Employee_Name"].str.strip()

# Replace missing salary with the median salary
median_salary = df["Salary"].median()
df["Salary"] = df["Salary"].fillna(median_salary)

# Convert Join_Date to datetime format
df["Join_Date"] = pd.to_datetime(df["Join_Date"])

# Create Years_of_Service column
df["Years_of_Service"] = 2026 - df["Join_Date"].dt.year

print("--- Preprocessed Data ---")
print(df)
```

Output:

```
--- Preprocessed Data ---
  Employee_Name   Department   Salary  Join_Date  Years_of_Service
0         Alice           HR  50000.0 2022-01-15                 4
1           Bob  Engineering  85000.0 2021-06-20                 5
2       Charlie           HR  72500.0 2023-03-11                 3
3         Diana  Engineering  92000.0 2020-11-01                 6
4          Evan    Marketing  60000.0 2024-02-28                 2
>>>
```

3. DATA AGGREGATION

Code:

```
File  Edit  Format  Run  Options  Window  Help
import pandas as pd
import numpy as np

# Employee Data
raw_data = {
    "Employee_Name": ["Alice", "Bob", "Charlie", "Diana", "Evan"],
    "Department": ["HR", "Engineering", "HR", "Engineering", "Marketing"],
    "Salary": [50000, 85000, 72500, 92000, 60000],
    "Join_Date": [
        "2022-01-15",
        "2021-06-20",
        "2023-03-11",
        "2020-11-01",
        "2024-02-28"
    ]
}

df = pd.DataFrame(raw_data)

# 3. DATA AGGREGATION
# Group by Department and calculate average salary and total employees
dept_summary = (
    df.groupby("Department")
      .agg(
          Avg_Salary=("Salary", "mean"),
          Total_Employees=("Employee_Name", "count")
      )
      .reset_index()
)

print("--- Aggregated Summary ---")
print(dept_summary)
```

Output:

```
--- Preprocessed Data ---
  Employee_Name   Department    Salary  Join_Date  Years_of_Service
0         Alice           HR  50000.0 2022-01-15                 4
1           Bob  Engineering  85000.0 2021-06-20                 5
2       Charlie           HR  72500.0 2023-03-11                 3
3         Diana  Engineering  92000.0 2020-11-01                 6
4          Evan    Marketing  60000.0 2024-02-28                 2
>>>
```

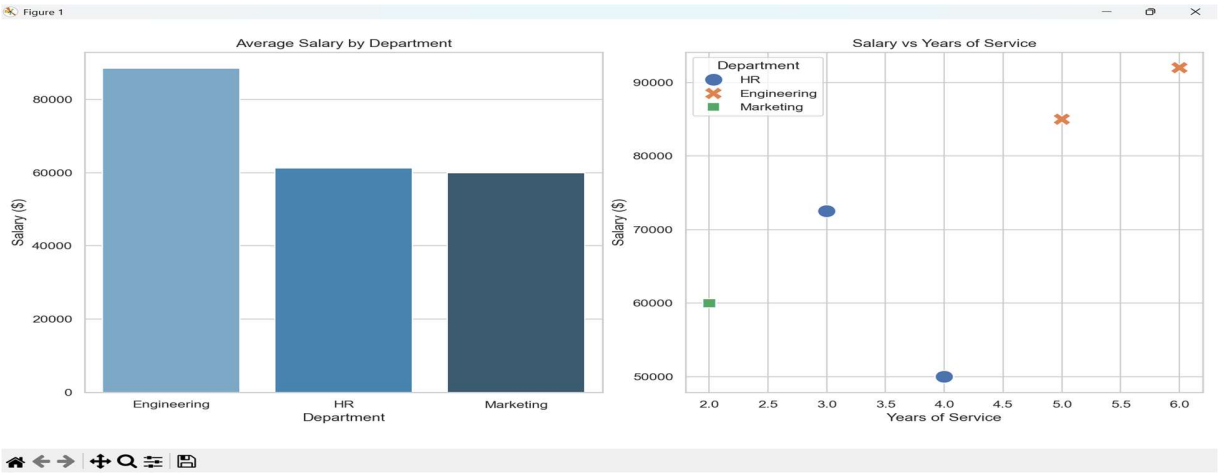
4. DATA VISUALIZATION

Code:

File Edit Format Run Options Window Help

```
        )
        .reset_index()
)

# 4. DATA VISUALIZATION

# Set theme
sns.set_theme(style="whitegrid")

# Create two plots
fig, axes = plt.subplots(1, 2, figsize=(14, 5))

# Plot 1: Bar Chart
sns.barplot(
    data=dept_summary,
    x="Department",
    y="Avg_Salary",
    hue="Department",
    palette="Blues_d",
    legend=False,
    ax=axes[0]
)

axes[0].set_title("Average Salary by Department")
axes[0].set_xlabel("Department")
axes[0].set_ylabel("Salary ($)")

# Plot 2: Scatter Plot
sns.scatterplot(
    data=df,
    x="Years_of_Service",
    y="Salary",
    hue="Department",
    style="Department",
    s=200,
    ax=axes[1]
)

axes[1].set_title("Salary vs Years of Service")
axes[1].set_xlabel("Years of Service")
axes[1].set_ylabel("Salary ($)")

plt.tight_layout()
plt.show()
```

Output:



Creating Sample Data Frame

Code:

```
File  Edit  Format  Run  Options  Window  Help
import pandas as pd

# Create an empty DataFrame
df = pd.DataFrame()

# Add columns
df['Name'] = ['John', 'Emma', 'Liam', 'Olivia']
df['Age'] = [20, 19, 21, 18]
df['Student'] = [True, True, False, True]

# Display the DataFrame
print(df)
```

Output:

```
    Name  Age  Student
0   John   20     True
1   Emma   19     True
2   Liam   21    False
3  Olivia  18     True
>>>
```

Adding Row to Data Frame

Code:

```
File  Edit  Format  Run  Options  Window  Help
import pandas as pd

# Create the DataFrame
df = pd.DataFrame()

df['Name'] = ['John', 'Emma', 'Liam', 'Olivia']
df['Age'] = [20, 19, 21, 18]
df['Student'] = [True, True, False, True]

# Create a new row
new_row = pd.DataFrame(
    [['Sophia', 22, False]],
    columns=['Name', 'Age', 'Student']
)

# Add the new row to the DataFrame
df = pd.concat([df, new_row], ignore_index=True)

# Display the updated DataFrame
print(df)
```

Output:

```
     Name  Age  Student
0    John   20     True
1    Emma   19     True
2    Liam   21    False
3   Olivia  18     True
4   Sophia  22    False
>>>
```

Data Analysis with SciPy

1.Importing Required Libraries

Code:

File Edit Format Run Options Window Help

```
import numpy as np
import pandas as pd
from scipy import stats

print("Libraries imported successfully.")
import numpy as np
from scipy import stats

data = [10, 20, 30, 40, 50]

print("Data:", data)
print("Mean:", np.mean(data))
print("Median:", np.median(data))
print("Mode:", stats.mode(data, keepdims=True))
```

2.Measures of Central Tendency Using SciPy

Code:

File Edit Format Run Options Window Help

```
import numpy as np
import pandas as pd
from scipy import stats

print("Libraries imported successfully.")
import numpy as np
from scipy import stats

data = [10, 20, 30, 40, 50]

print("Data:", data)
print("Mean:", np.mean(data))
print("Median:", np.median(data))
print("Mode:", stats.mode(data, keepdims=True))
```

3.Probability Distribution Analysis Using SciPy

Code:

File Edit Format Run Options Window Help

```
from scipy.stats import norm

probability = norm.cdf(85, loc=70, scale=10)

print("Probability:", round(probability, 4))
```

4.Hypothesis Testing

Code:

File Edit Format Run Options Window Help

```
from scipy import stats

data = [22, 25, 19, 24, 28, 30]

t_stat, p_value = stats.ttest_1samp(data, 25)

print("T-Statistic:", round(t_stat, 3))
print("P-Value:", round(p_value, 3))
```

5.Correlation Analysis

Code:

```
Edit  Format  Run  Options  Window  Help
from scipy.stats import pearsonr

x = [1, 2, 3, 4, 5]
y = [2, 4, 6, 8, 10]

corr, p_value = pearsonr(x, y)

print("Correlation:", corr)
print("P-Value:", p_value)
```

6. Linear Algebra Operations

Code:

```
File  Edit  Format  Run  Options  Window  Help
from scipy import linalg

A = [[3, 2],
     [1, 2]]

B = [5, 5]

solution = linalg.solve(A, B)

print("Solution:", solution)
```

7.Optimization Using SciPy

Code:

```
File  Edit  Format  Run  Options  Window  Help
from scipy.optimize import minimize

def objective(x):
    return x**2 + 4

result = minimize(objective, x0=5)

print("Optimal Value of x:", result.x)
print("Minimum Function Value:", result.fun)
```

Output:

```
File  Edit  Shell  Debug  Options  Window  Help
Python 3.12.3 (tags/v3.12.3:f6650f9, Apr  9 2024, 14:05:25) [MSC v.1938 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/addinglibraries.py
Libraries imported successfully.
>>>
= RESTART: C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/addinglibraries.py
Libraries imported successfully.
Data: [10, 20, 30, 40, 50]
Mean: 30.0
Median: 30.0
Mode: ModeResult(mode=array([10]), count=array([1]))
>>>
= RESTART: C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/pda using scipy.py
Probability: 0.9332
>>>
= RESTART: C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/hypothesistesting.py
T-Statistic: -0.205
P-Value: 0.846
>>>
= RESTART: C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/correlational analysis.py
Correlation: 1.0
P-Value: 0.0
>>>
= RESTART: C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/Linear operation.py
Solution: [0.  2.5]
>>>
= RESTART: C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/optimization using scipy.py
Optimal Value of x: [-2.62955131e-08]
Minimum Function Value: 4.000000000000001
>>>
```

Scikit Library in Python

1. Logistic Regression (Supervised Learning)

Code:

```
File  Edit  Format  Run  Options  Window  Help
import numpy as np
from sklearn import datasets
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score

# Load Iris dataset
iris = datasets.load_iris()
X = iris.data
y = iris.target

# Split dataset into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)

# Standardize the data
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

# Train Logistic Regression model
log_reg = LogisticRegression()

log_reg.fit(X_train, y_train)

# Predict test data
y_pred = log_reg.predict(X_test)

# Calculate accuracy
accuracy = accuracy_score(y_test, y_pred)

print("Accuracy:", accuracy)
```

Output:

```
Accuracy: 1.0
>>>
```

2. K Means Clustering (Unsupervised Learning)

Code:

```
File  Edit  Format  Run  Options  Window  Help
from sklearn.datasets import load_iris
from sklearn.cluster import KMeans

# Load Iris dataset
iris = load_iris()

# Create KMeans model
kmeans = KMeans(n_clusters=3, random_state=42, n_init=10)

# Train the model
kmeans.fit(iris.data)

# Get cluster labels
cluster_labels = kmeans.labels_

print("Cluster Labels:")
print(cluster_labels)
```

Output:

```
Cluster Labels:
[1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
 1 1 1 1 1 1 1 1 1 1 1 1 1 0 0 2 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
 0 0 0 2 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 2 0 2 2 2 2 0 2 2 2 2
 2 2 0 0 2 2 2 2 0 2 0 2 0 2 2 0 0 2 2 2 2 2 0 2 2 2 2 0 2 2 2 0 2 2 2 0 2
 2 0]
>>>
```